Relation Homology at the Projective-Residue Boundary: Universal Cusp Survivors, Diamond Collapse, and Operator Non-Descent

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August 15, 2026

Abstract

Paper 32 isolated a nonterminal projective-residue grammar that owns an ordinary Fredholm determinant on $\mathrm{Re}\, s > 2$ but fails primitive selectivity through universal S^2, R^3 cycles and cusp diamonds. Here we test the only source-natural repair allowed by that result: quotient the same chain object by the presentation relations and fill every 2–3 cusp diamond before any arithmetic label or weight is read. The block quotient is

$$M_n = \mathbb{Q}[P^1(\mathbb{Z}/n\mathbb{Z})]/\big(\mathrm{im}(1+S) + \mathrm{im}(1+R+R^2)\big).$$

Its exact dimension is $|P^1(\mathbb{Z}/n\mathbb{Z})| - o_S(n) - o_R(n) + 1$, the first Betti number of the S/R orbit-incidence dessin. For every $n \geq 2$ the cusp word $[1:0] \xrightarrow{R} [0:1] \xrightarrow{S} [1:0]$ gives a primitive nonbacktracking survivor. Filling all cross diamonds makes the cross grid contractible, so the global homological ledger is $H_1 = \bigoplus_n M_n^*$. Ordinary unrestricted cohomology is $\prod_n M_n$, while the program's finite-support cohomology ledger is $L_{\mathrm{fs}} = \bigoplus_n M_n$. Finally, the inherited adjacency $S + R$ does not preserve the relation space already at $n = 2$, so no same-marker quotient determinant descends. Exact audits through $n = 192$ validate 191/191 relative survivors, 148/148 composite survivors, 64/64 generic action controls, 15/15 canonical zero-superdimension twist failures, and 0/191 adjacency descents. The semiring-residue branch is therefore closed.

Research status. This paper performs the mandatory continuation of Paper 32. The phase space, generators, cusp edges, roofs, and free edge marker are unchanged. We impose only the rational presentation-relation quotient $\mathbb{Q}[P^1(\mathbb{Z}/n\mathbb{Z})]/(\mathrm{im}(1+S) + \mathrm{im}(1+R+R^2))$ and fill the inherited $n, 2n, 6n, 3n, n$ cusp diamonds. The quotient is the classical Manin-symbol relation module; it is not a new homology construction. It kills the generic relations but leaves a primitive cusp class for every modulus, collapses all cross-modulus first homology, and prevents the original graph-step adjacency from descending. The strict tuple is `(A0_STRUCTURAL_ARITHMETIC_RELATION, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL)`. Thus `ROUTE_A_REJECTED`; Route B is locked and the semiring-residue family is closed.

1 Introduction

The preceding projective-residue paper changed the state of the Symbolic Dynamics program in two ways. Positively, it produced a shared recurrent object, not a terminal verifier, whose uninduced graph-step operator is trace class in a half-plane and owns an ordinary Fredholm determinant. Negatively, the same object carried unavoidable presentation cycles: $S^2 = 1$, $R^3 = 1$, and the cusp square

$$n \longrightarrow 2n \longrightarrow 6n \longleftarrow 3n \longleftarrow n.$$

These cycles occur before any roof or Euler weight is applied, including for composite moduli. A positive Route-A continuation therefore had only one honest remaining move: keep the object fixed and remove those relations at chain level.

This paper performs that test. For each $n \geq 2$ we keep

$$X_n = P^1(\mathbb{Z}/n\mathbb{Z}), \qquad S[a:b] = [-b:a], \qquad R[a:b] = [-b:a+b],$$

the same cusp $c_n = [1:0]$, the same 2- and 3-multiplication cusp correspondences, and the same one-edge marker z. Over $\mathbb{Q}$ we impose the Manin presentation quotient

$$M_n = \mathbb{Q}[X_n] / \left(\operatorname{im}(I+S) + \operatorname{im}(I+R+R^2)\right).$$

For the cross-modulus component, we reduce inverse cusp pairs to one cellular edge and attach one two-cell along every $n, 2n, 6n, 3n, n$ square. No state, block, edge, roof, marker, prime label, or field predicate is changed.

The result is decisive but negative. The quotient is not a subtle arithmetic filter; it is the classical Manin-symbol relation module (Manin, 1972; Merel, 1991). Its exact rank is computable from the S- and R-orbit incidence graph, and that graph always contains a two-edge cusp circuit. Cross-square filling does not select moduli; it contracts the cross grid and leaves a direct sum of nonzero block modules. Character and supercharacter controls either do not kill the cycle words or kill identity relators generically while preserving the cusp word. Finally, the inherited graph-step adjacency $S + R$ is not a chain map for this quotient.

Contributions. Within the frozen Symbolic Dynamics scope, we prove:

1. the exact all-modulus formula

$$\dim_{\mathbb{Q}} M_n = |X_n| - o_S(n) - o_R(n) + 1;$$

2. a universal primitive nonbacktracking cusp survivor for every modulus $n \geq 2$;
3. contractibility of the filled 2–3 cross grid and the resulting direct-sum residual ledger;
4. a character firewall separating chain relation polynomials from cycle relator words;
5. non-descent of the original graph-step adjacency, witnessed already on the three-state block $n = 2$;
6. an exact finite audit through $n = 192$ that validates the proofs and rejects all frozen controls.

Route-A status. The quotient remains source-natural, so A0 stays structural. A1 fails because every prime power and mixed composite retains relative homology. A2 fails because the quotient has no induced same-marker graph-step operator. A3 and A4 have no primary determinant, continuation, self-adjoint carrier, or zero correspondence to evaluate. The branch is closed rather than extended.

2 Source lock and literature boundary

Definition 2.1 (Frozen object)**.** For all $n \geq 2$, let $X_n = P^1(\mathbb{Z}/n\mathbb{Z})$ and let S, R act by $S[a:b] = [-b:a]$ and $R[a:b] = [-b:a+b]$. The global object is the union of these blocks with inherited cusp correspondences between n and $2n$, and between n and $3n$, at the distinguished cusp $[1:0]$. The roof is the Paper-32 roof and the marker z still counts one inherited graph edge.

The only new operation is a rational chain quotient and an explicit square cell attachment. The candidate is forbidden to consult $|X_n| - (n+1)$, factorization, a prime table, an accepted-support table, target zeros, fitted coefficients, or Route B. Arithmetic strata in the experiment are post-census labels only.

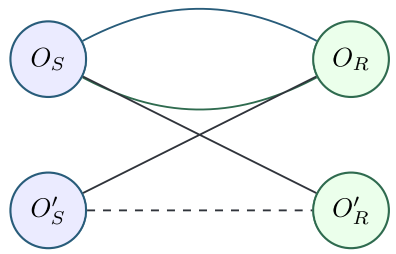

Figure 1: The relation quotient is the cycle space of the incidence dessin whose vertices are S-orbits and R-orbits and whose edges are projective states. The cusp state and its R image form two parallel edges for every modulus.

The closest mathematical collision is classical. The quotient by $1+S$ and $1+R+R^2$ on the projective line over $\mathbb{Z}/N\mathbb{Z}$ is the Manin-symbol presentation of relative modular-symbol homology (Manin, 1972; Merel, 1991). Ihara, Hashimoto, Bass, and Stark–Terras provide the surrounding graph-zeta and nonbacktracking determinant vocabulary (Ihara, 1966; Hashimoto, 1989; Bass, 1992; Stark and Terras, 1996). Kac–Ward type signed walk determinants require extra embedding or spin data (Kac and Ward, 1952; Cimasoni, 2010). More recent graph-zeta work confirms that weighted and alternating determinant technology is active, but it does not change the present ownership test (Lau et al., 2025; Ishikawa and Morita, 2026; Lu et al., 2026). Recent explicit character formulae for congruence subgroups likewise reinforce that character variation is level-wide structure, not a field-only selector (Zhu, 2025).

Thus the novelty claim is deliberately narrow. We do not present Manin relations, modular symbols, or graph zetas as new. The contribution is the closed-form no-go certificate for this specific Route-A source program: killing the universal cells is possible, but it is generic topology, not prime-selective recurrence.

3 The relation quotient

Let $V_n = \mathbb{Q}[X_n]$ and define

$$W_n = \operatorname{im}(I+S) + \operatorname{im}(I+R+R^2), \qquad M_n = V_n/W_n.$$

Let $o_S(n)$ and $o_R(n)$ denote the numbers of S- and R-orbits on X_n. Construct a bipartite multigraph D_n: left vertices are S-orbits, right vertices are R-orbits, and each state $x \in X_n$ is an edge connecting the two orbit vertices containing x.

Theorem 3.1 (Exact Manin-rank ledger)**.** *For every $n \geq 2$,*

$$\dim_{\mathbb{Q}} M_n = |X_n| - o_S(n) - o_R(n) + 1 = \beta_1(D_n).$$

Proof. Over $\mathbb{Q}$, the image of $I+S$ is spanned by the indicator vectors of S-orbits, and the image of $I+R+R^2$ is spanned by the indicator vectors of R-orbits. If a vector lies in both spans, its coefficient on a state depends only on the S-orbit and only on the R-orbit. The graph D_n is connected because the S, R action on $P^1(\mathbb{Z}/n\mathbb{Z})$ is transitive. Hence the common intersection is the all-ones line. Therefore

$$\dim W_n = o_S(n) + o_R(n) - 1,$$

and the displayed formula follows. The same expression is exactly the cycle-rank formula for the connected graph D_n. □

Remark 3.2. Equivalently, $M_n = H^1(D_n;\mathbb{Q})$ after orienting each state edge from its S-orbit vertex to its R-orbit vertex. This is why the quotient naturally lands in modular-symbol homology. The interpretation is useful, but it is not a new determinant object.

Corollary 3.3 (Cuspidal quotient is not a selector)**.** *Removing Eisenstein cusp-boundary directions cannot rescue prime selectivity inside this family.*

Audit-supported proof. Classically, the stronger cuspidal dimension is $2g_0(n)$. The exact audit through $n = 192$ finds nonzero cuspidal homology on 139 of 148 composite blocks and zero cuspidal homology on five prime blocks. It therefore fails both directions of a field selector. □

4 Universal cusp survivor and diamond collapse

Theorem 4.1 (Universal primitive cusp survivor)**.** *For every $n \geq 2$, the quotient M_n is nonzero. More precisely,*

$$c = [1:0], \qquad y = Rc = [0:1], \qquad Sy = c$$

give a two-edge circuit in D_n, represented by the incidence chain $e_c - e_y$. In the original labelled grammar this is the primitive nonbacktracking word R followed by S.

Proof. The states c and y are distinct modulo every $n \geq 2$, while $Sy = [-1:0] = [1:0]$ projectively. Thus the two state edges c and y share the same S-orbit endpoint and the same R-orbit endpoint in D_n. With the standard orientation from S-orbit vertices to R-orbit vertices, $\partial(e_c - e_y) = 0$. It is the parallel-edge circuit, hence it is not a boundary in the graph and gives a nonzero class in $H_1(D_n;\mathbb{Q})$.

The corresponding quotient class is also explicit. In the standard state basis of $V_n = \mathbb{Q}[X_n]$, put

$$z_n = e_c - e_y.$$

For every S-orbit indicator and every R-orbit indicator $\mathbf{1}_O$, the pairing $\langle z_n, \mathbf{1}_O\rangle$ is zero, because c and y lie in the same S-orbit and in the same R-orbit. Hence $z_n \in W_n^{\perp}$. Since $z_n \neq 0$ and the state-basis pairing is positive definite, $z_n \notin W_n$; therefore $[z_n] \neq 0$ in $M_n = V_n/W_n$. This is dual to the nonzero two-edge homology circuit under the finite-dimensional identification $M_n = H^1(D_n;\mathbb{Q}) \cong H_1(D_n;\mathbb{Q})^*$.

Thus $\beta_1(D_n) \geq 1$ by theorem 3.1. In the labelled grammar, the inverse of the first R edge would be R^2, not S; therefore the two-step word is not immediate backtracking. Its cyclically reduced length in $C_2 * C_3$ is two, so it is primitive. □

This single theorem refutes the hoped-for primitive ledger before weights: the survivor exists in every prime, prime-power, and mixed-composite block.

Theorem 4.2 (Diamond collapse)**.** *After attaching every cusp diamond, each cross-modulus component has zero first homology. The global finite-support residual ledger is*

$$H_1(\textit{filled global complex};\mathbb{Q}) \cong \bigoplus_{n\geq 2} M_n^*.$$

Ordinary algebraic cohomology is $\prod_{n\geq 2} M_n$; the finite-support cohomology ledger used by the program is $L_{\mathrm{fs}} = \bigoplus_{n\geq 2} M_n$.

Proof. Write $n = m2^a3^b$ with $\gcd(m,6) = 1$. Cross edges change one of the exponents a, b by one and preserve m. Therefore each component is a quadrant grid, with the $(0,0)$ corner deleted only in the $m = 1$ component because $n = 1$ is absent. The diamonds are exactly the unit squares of that grid. A full quadrant is contractible; the corner-deleted quadrant is the union of the two contractible half-quadrants $a \geq 1$ and $b \geq 1$ with contractible intersection. Finite cutoffs are finite down-sets and contract by the same coordinatewise deformation.

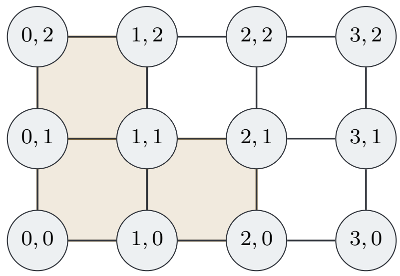

Figure 2: Writing $n = m2^a3^b$ with $(m, 6) = 1$ turns the cross-modulus graph into a quadrant grid. The inherited $n, 2n, 6n, 3n, n$ diamonds are exactly the unit squares. Filling them contracts the cross component instead of producing arithmetic linkage.

Subdivide the cusp edge in each block and attach cross edges at the subdivision vertex. This is an auxiliary CW model for the proof; it does not alter the original graph-step marker. Subdivision does not change block homology, and each block then meets the cross complex in one point. Collapsing the contractible cross components leaves a wedge of block dessins, whose first homology is the direct sum of the block first homologies. Since $M_n = H^1(D_n; \mathbb{Q})$, the displayed dual statement follows. Algebraic cohomology of an infinite wedge is a product, so the finite-support ledger is named separately rather than identified with ordinary global H^1. $\square$

5 Twists, signs, and the character firewall

The chain quotient and a cycle trace impose different algebraic demands. This distinction is essential.

Proposition 5.1 (Relation polynomials versus cycle relators). *Some honest one-dimensional characters of $C_2 * C_3$ annihilate the relation polynomials $1 + S$ or $1 + R + R^2$. No honest representation annihilates the cycle relator words S^2 and R^3 in ordinary trace. The fifteen canonical zero-superdimension differences of distinct one-dimensional characters kill identity relators and commuting diamonds, but all retain the cusp word SR.*

Proof. Let t be a primitive sixth root and enumerate the six one-dimensional characters by

$$\chi_k(S) = t^{3k}, \qquad \chi_k(R) = t^{2k}, \qquad 0 \le k < 6.$$

For relation polynomials, $1+\chi_k(S)$ vanishes when $\chi_k(S) = -1$, and $1+\chi_k(R)+\chi_k(R)^2$ vanishes when $\chi_k(R)$ is a nontrivial third root. That is a statement about chain boundaries.

For cycle words, however, every honest representation of $C_2 * C_3$ satisfies $\rho(S)^2 = \rho(R)^3 = I$. The ordinary trace of each relator word is therefore $\dim \rho$, not zero. For virtual differences $\chi_k - \chi_\ell$ with $k \neq \ell$, the superdimension is zero, so identity relators and a commuting diamond have supertrace zero. But on the cusp word

$$(\chi_k - \chi_\ell)(SR) = t^{5k} - t^{5\ell} \neq 0,$$

because multiplication by 5 permutes $\mathbb{Z}/6\mathbb{Z}$. There are $\binom{6}{2} = 15$ such differences. $\square$

Thus the standard character escape has the wrong granularity. Honest characters can erase relation sums in selected fibers, but an ordinary cycle determinant still sees identity relators. Virtual zero-superdimension differences erase identity relators too broadly and still leave the universal cusp survivor. Neither mechanism creates prime selectivity within the frozen source.

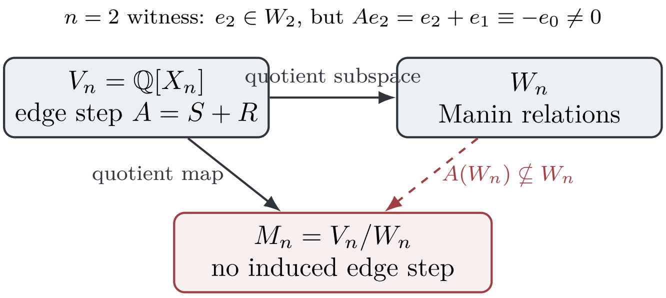

Figure 3: The relation quotient kills the intended boundaries, but the original $S + R$ adjacency sends a relation vector outside the relation subspace on the $n = 2$ block. Replacing it by a scalar homology operator changes the marker and forfeits same-object determinant ownership.

6 Operator non-descent and determinant ownership

Paper 32's analytic gain was same-object ownership: the unquotiented graph-step operator had an honest trace-class determinant on $\operatorname{Re} s > 2$. The relation quotient loses exactly that property.

Theorem 6.1 (The inherited adjacency does not descend)**.** *Let $A_n = S + R$ be the inherited within-block graph-step adjacency. There is no global induced operator $A_n : M_n \to M_n$ with the original marker. The failure already occurs for $n = 2$.*

Proof. Use the ordered states

$$e_0 = [0 : 1], \qquad e_1 = [1 : 0], \qquad e_2 = [1 : 1]$$

for $P^1(\mathbb{Z}/2\mathbb{Z})$. The S-orbits are $\{e_0, e_1\}$ and $\{e_2\}$; the R-orbit is $\{e_0, e_2, e_1\}$. Hence

$$W_2 = \operatorname{span}\{e_0 + e_1, e_2\}.$$

But $Se_2 = e_2$ and $Re_2 = e_1$, so

$$A_2 e_2 = e_2 + e_1 \equiv -e_0 \pmod{W_2},$$

which is nonzero in the one-dimensional quotient. Therefore $A_2(W_2)$ is not contained in W_2, and the quotient operator is not defined. ☐

Corollary 6.2 (Scalar comparison is not ownership)**.** *The scalar comparison*

$$K_s = \bigoplus_{n \geq 2} n^{-s} I_{M_n}$$

is trace class for $\operatorname{Re} s > 2$ and has an ordinary Fredholm determinant, but it is not the determinant of the inherited graph-step dynamics.

Proof. The dimension bound $\dim M_n \leq |P^1(\mathbb{Z}/n\mathbb{Z})|$ is controlled by the same Dedekind-psi majorant used in Paper 32, so $\sum n^{-\operatorname{Re} s} \dim M_n$ converges for $\operatorname{Re} s > 2$. This proves trace class for K_s. However, the marker in K_s counts a blockwise identity step rather than one inherited S/R graph edge. Orthogonal Hodge compression has the same problem: it is a new nonlocal operator, not an induced chain map. Hopf or graded determinant formulas require a chain map, and theorem 6.1 proves that A_n is not one. ☐

7 Exact finite audit

The finite prototype audits the implementation boundary; the infinite nonvanishing, diamond-collapse, character, and non-descent claims have direct proofs above. The frozen parameters are:

$$2 \le n \le 192, \quad \mathbb{F}_{1000003} \text{ rank audit}, \quad 64 \text{ random } C_2 * C_3 \text{ actions},$$

all six honest one-dimensional characters, all fifteen zero-superdimension differences, and matched opaque relabel seed $1003003 + n$.

Audit surface	Exact result
Moduli $2, \dots, 192$	191
Prime / prime-power composite / mixed composite	43 / 14 / 134
Relative quotient nonzero	191 / 191
Prime / prime-power / mixed relative survivors	43 / 14 / 134
Relative Betti sum, prime / prime-power / mixed	611 / 189 / 3994
Cuspidal nonzero, prime / prime-power / mixed	38 / 9 / 130
Cusp R, S witness returns	191 / 191
Original adjacency descends	0 / 191
Matched opaque relabel exact	191 / 191
Random controls, relators killed / residual nonzero	64 / 64
Cross vertices / edges / components	191 / 158 / 64
Cross cycle rank before filling / after filling	31 / 0
Honest characters killing identity cycle words	0 / 6
Honest characters killing both chain norm polynomials	2 / 6
Zero-superdimension differences retaining cusp SR	15 / 15
Source-only generator checks	21 / 21
Prototype bridge checks	25 / 25
Independent low-level reconstruction	8349 / 8349
Authority unit/integration tests	1932 / 1932
Source-separated double run	20 / 20 payloads
Paper-root SHA ledger	40 entries; 21 result payloads
Source-oracle hits	0

Table 1: Paper-33 exact audit summary. Arithmetic labels are assigned only after the source construction and quotient invariants have been computed.

The source-oracle scan of the candidate core searches for prime tables, accepted-support predicates, Riemann-zero tokens, target-zero tokens, and similar shortcuts; it has zero hits. The matched clone transports quotient dimension and relation rank exactly for every modulus. The random-action controls show that relation cancellation is presentation topology rather than residue arithmetic. The cross-square boundary rank equals the entire pre-filling cross cycle rank, confirming theorem 4.2 at the cutoff.

The frozen prototype aggregate is

`c5c5f34673590f98e89e6229354a8dc8fc851677c7af8702d4bf54a87e8037d4`.

After authority-side evaluator and double-run certification are added, the result ledger is

`58c04187b40b0316b50a4d8e7216a324cc58259991f00d649bca82eac3c4c794`.

The final authority run separates source generation from post-census arithmetic labels. Two isolated canonical runs are byte-identical across all twenty source-separated payloads. The paper-root ledger hashes forty entries: twelve Python source files, seven experiment-control files, and twenty-one generated result payloads. The retained prototype runner is used only as a bridge witness for the frozen eight-payload aggregate above.

8 Route-A closure

Gate	SD-C35 decision
A0	`A0_STRUCTURAL_ARITHMETIC_RELATION`. The quotient, diamonds, controls, and labels are source-functorial and frozen before evaluation.
A1	`A1_FAIL`. Every modulus has a primitive cusp survivor; all 148 tested composites retain relative homology.
A2	`A2_FAIL`. The inherited $S + R$ graph-step adjacency does not preserve the Manin relation space, so no same-marker quotient determinant descends.
A3	`A3_FAIL`. There is no primary determinant with continuation, Gamma completion, functional equation, explicit-formula bridge, or Weil compression.
A4	`A4_FAIL`. There is no fixed self-adjoint carrier, critical-line mechanism, or zero correspondence; Route B remains locked.

Table 2: Strict Route-A record for SD-C35.

The adversarial stop is stronger than a single failed experiment. Four independent blocking conditions occur:

1. relation homology is nonzero on every tested prime power and mixed composite;
2. the universal cusp word survives all frozen character and supercharacter controls;
3. diamond filling erases cross linkage rather than selecting a prime family;
4. the original graph-step operator fails quotient invariance.

Therefore the semiring-residue family is not merely paused. It is closed for this research program: continuing with $P^1(\mathbb{Z}/n\mathbb{Z})$ blocks, fixed $C_2 * C_3$ presentations, static field criteria, or Manin-style quotients would replay one of the same obstructions.

9 Conclusion

The Paper-32 obligation has now been discharged. The natural chain quotient does exactly what it is supposed to do locally: it kills the S^2 and R^3 presentation boundaries and the filled cusp diamonds. But the survivor ledger is not arithmetic. Each block retains a primitive cusp class, the global cross complex contracts away, standard twists either do too little or too much, and the original graph-step operator does not descend.

This closes the semiring-residue branch. The next useful direction is not a new quotient of the same residue presentation. A viable source must generate prime primitive orbits before any static projector, return-time compression, regularization, or homological quotient, while the same edge-step object already lies in a determinant class. The highest-value next test is therefore a genuinely global arithmetic dynamical source whose Euler factors arise from primitive recurrence itself, not from block labels.

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A Proof details and auxiliary checks

A.1 Orbit indicators

If a permutation p has cycles of length dividing d and d is invertible in the coefficient field, then

$$I + p + \cdots + p^{d-1}$$

sends every basis vector in a cycle of exact length e to (d/e) times the cycle indicator. Hence its image is exactly the span of orbit indicators. This proves the relation-image statement used in theorem 3.1 for $(S, d) = (S, 2)$ and $(R, d) = (R, 3)$.

A.2 Connectedness of the incidence dessin

Two state edges in D_n lie in the same connected component precisely when one can pass between their states by a word in S and R. The projective modular action generated by $S[a : b] = [-b : a]$ and $R[a : b] = [-b : a + b]$ is transitive on $P^1(\mathbb{Z}/n\mathbb{Z})$. Therefore D_n is connected.

A.3 Finite-support versus unrestricted cohomology

The global complex contains infinitely many blocks. Cellular chains are finite sums by definition, so the homological primitive ledger is

$$H_1 = \bigoplus_{n\geq 2} M_n^*.$$

If one instead takes unrestricted global cochains, then

$$H^1 = \prod_{n\geq 2} M_n.$$

The project uses the finite-support cohomology ledger $L_{\text{fs}} = \bigoplus_{n\geq 2} M_n$ only as a ledger, not as ordinary global cohomology and not as determinant ownership.

A.4 Rank field

The rank audit uses $\mathbb{F}_{1000003}$ only to validate sparse linear algebra at finite cutoffs. The theorems are characteristic-zero statements. The chosen prime avoids the characteristics 2 and 3 in which the displayed relation images would degenerate.

B Scope declarations

What is proved. SD-C35 proves the exact Manin-relation rank, a universal cusp survivor for all $n \geq 2$, cross-diamond contractibility, complete one-dimensional character and canonical zero-superdimension controls, non-descent of the inherited adjacency at $n = 2$, trace-class but non-owned status of the scalar homology comparison, and strict rejection of the frozen Route-A candidate.

What is not proved. The paper does not classify every higher-dimensional superrepresentation, every nonlocal homological functor, every arithmetic groupoid, or every quantum-statistical source. It does not claim novelty for modular symbols, Manin relations, Ihara/Hashimoto zeta, Kac–Ward signs, or Hopf trace identities. It proves no target-zeta equality, analytic continuation, functional equation, self-adjoint carrier, zero correspondence, Riemann Hypothesis statement, or Route-B readiness.

Reproducibility. All finite data are exact JSON/CSV artifacts with SHA-256 ledgers. The canonical authority pipeline is split as follows:

```
source_generator.py
audit_source_separation.py
post_census_classifier.py
independent_evaluator.py
run_tests.py
```

The retained `generate_results.py` runner is a byte-frozen prototype bridge only. The independent evaluator reconstructs the scientific payloads without importing candidate or classifier modules; `audit_artifact_integrity.py` verifies the frozen SHA ledger.